

Habitat & Cultivation Found in the wild throughout Europe, Asia, and Australia, sandwort thrives in sandy and gravelly places, especially close to the sea.

Parts Used Aerial parts.

Medicinal Actions & Uses Sandwort is a diuretic herb that is thought to relax the muscle walls of the urinary tubules and bladder. Sandwort is most commonly taken in the form of an infusion to treat kidney stones, acute and chronic cystitis, and other conditions of the bladder.

Related Species Seabeach sandwort (*A. peploides*), a closely related northern plant, is eaten by the Inuit of Alaska as a fresh, pickled, or oil-preserved vegetable. In Iceland, this plant is fermented and eaten in the same way as sauerkraut. Rupturewort (*Herniaria glabra*, p. 219), which is a European plant, has medicinal properties that are similar to those of sandwort.

Argemone mexicana

(Papaveraceae)

Mexican Poppy, Chicalote

Description Prickly annual growing to 3 ft (1 m). Has spiny white-veined leaves and large yellow flowers with delicate petals.

Habitat & Cultivation Mexican poppy grows in tropical regions from the southernmost U.S. to South America. It favors dry soil and is often found in tobacco fields.

Parts Used Aerial parts, latex, seeds.

Constituents Mexican poppy contains isoquinoline alkaloids similar to those in the opium poppy (*Papaver somniferum*, p. 244).

History & Folklore Like most poppies, this plant exudes a milky latex, which was traditionally used in Ecuador to treat cataracts.

Medicinal Actions & Uses Commonly used through much of Latin America, the whole plant acts as a mild painkiller and sedative, and is taken



Mexican poppy flowers have expectorant properties and are useful for treating coughs.

to relieve muscle ache, toothache, and neuralgic pain. The fresh latex contains protein-dissolving properties and is traditionally applied to warts and cold sores. The flowers and seeds are mildly expectorant and can be used to treat irritable coughs and asthma.

Related Species In Hawaii, the latex of *A. glauca* is also used to treat warts.

⚠ Cautions Excessive use can cause glaucoma. Use only under professional supervision.

Arisaema spp. (Araceae)

Tian Nan Xing

Description Perennial herb growing to 3 ft (1 m). Has star-shaped leaves and purple-white or green pitcher-like bracts.

Habitat & Cultivation *Tian nan xing* grows wild in eastern Asia, especially in China, where it is widely cultivated. The rhizome is unearthed in autumn or winter.

Part Used Dried rhizome.

Constituents *Tian nan xing* contains triterpenoid saponins and benzoic acid.

Medicinal Actions & Uses In Chinese herbal medicine, *tian nan xing* is thought to encourage the coughing up of phlegm. The dried rhizome is used principally for chest problems. When prescribed internally it is always combined with fresh ginger root (*Zingiber officinale*, p. 155). The fresh rhizome is only ever used externally, for ulcers and other skin conditions.

Related Species Jack-in-the-pulpit or Indian turnip (*A. triphyllum*), a North American species, is a treatment for chest conditions.

⚠ ⚠ Cautions Take only under professional supervision. The fresh rhizome is very toxic; use only the dried rhizome internally.

Aristolochia clematitis

(Aristolochiaceae)

Birthwort

Description Unpleasant-smelling perennial with heart-shaped leaves and tubular yellow flowers with flattened lips.

Habitat & Cultivation Native to central and southern Europe, birthwort is also found in southwestern Asia. The root is unearthed in spring or autumn.

Parts Used Root, aerial parts.

Constituents Birthwort contains aristolochic acids, a volatile oil, and tannins. While stimulating white blood cell activity, aristolochic acid is carcinogenic and toxic to the kidneys (see Research).

History & Folklore *Aristolochia* means "excellent birth," and refers to the traditional use

of the fresh juice to induce labor. Theophrastus (c. 372–286 BCE) records that the plant was used to treat disorders of the uterus, reptile bites, and sores on the head.

Medicinal Actions & Uses No longer in use today and banned in Europe and North America, birthwort was formerly used to treat wounds, sores, and snake bite. It has been taken after childbirth to prevent infection and is also a potent menstruation-inducing herb and a (very dangerous) abortifacient. A decoction was taken to encourage healing of ulcers. Birthwort has also been used for asthma and bronchitis.

Research Birthwort illustrates the fact that though a plant is natural, this has no bearing on its safety. Aristolochic acid (present within birthwort) is a kidney toxin that induces kidney failure and cancer within the kidneys and urinary tract. However, the slow rate of development of symptoms has meant that in the past no connection was made between these kidney symptoms and the herb. These toxic effects apply to birthwort and all *Aristolochia* species that contain aristolochic acid. Kidney failure and death from aristolochic acid is relatively rare in the West but is "reaching potentially epidemic proportions in the East," according to research published in Taiwan in 2013. Many species of *Aristolochia* continue to be used in Oriental herbal medicine in countries such as China, Japan, and Taiwan. All species of *Aristolochia* are banned in most Western countries.

⚠ Caution Under no circumstance use birthwort or any *Aristolochia* species as medicines.

Armoracia rusticana syn.

Colchlearia armoracia

(Brassicaceae)

Horseradish

Description Perennial with a deep tap root, leaves up to 20 in (50 cm) long, and clusters of 4-petaled white flowers.

Habitat & Cultivation Native to Europe and western Asia, this herb is widely cultivated for its root, unearthed in autumn.

Parts Used Root, leaves.

Constituents Horseradish root contains glucosilates (mainly sinigrin), flavonoids, asparagine, resin, and vitamin C. On being crushed, sinigrin produces allyl isothiocyanate, an antibiotic substance. The flavonoids have been shown to be antioxidant.

History & Folklore Pliny (23–79 CE) probably had horseradish in mind when describing a plant that warded off scorpions, but for most of its long history, horseradish has been used mainly as a diuretic herb. It is a popular condiment, particularly in Britain and central Europe.